BRAVO Lab

Brazilian Risk, Allocation, Volatility & Options Lab

Institutional Executive Report

PRIVATE APPLICATION DRAFT · v0.2.0

A reproducible Python research prototype that turns Brazilian market stress, derivatives overlay logic, active-risk diagnostics, and stress transmission signals into decision-grade portfolio-governance material.

CREATED BY

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Signal date: 2026-06-09 · Not investment advice

BRAVO

SCENARIO

PROJECT TYPE	Portfolio-governance research prototype
REGION	Brazil / B3 / Cross-asset stress transmission
ASSET LAYER	Brazil equity beta, derivatives overlays, volatility, active risk, stress signals
CURRENT SIGNAL DATE	2026-06-09
CURRENT REGIME	n/a
DOMINANT CHANNEL	n/a
CONTEXT	BRAVO Lab maps Brazilian market stress into portfolio decisions: passive beta retention, covered-call income, collar protection, or BSTI policy switching.
OUTPUT	Institutional executive PDF, reproducible evidence files, charts, CSV outputs, and audit-ready code.

EXECUTIVE SIGNAL

The decision layer matters more than the model output

BRAVO Lab should not be presented as a market-prediction machine. Its strongest use is more institutional: it turns stress evidence into a governed portfolio discussion about beta exposure, volatility income, protection cost, and implementation risk.

BSTI SCORE	REGIME	POLICY
n/a	n/a	n/a

Decision implication

The current action is not a directional call. It is an overlay discussion triggered by stress evidence. A covered-call read implies income capture may be acceptable only if the portfolio committee is comfortable selling part of a recovery.

Risk committee question

Is the stress state persistent enough to justify action, or is it only a temporary pressure print? Would the portfolio team prefer income discipline, downside protection, or full beta retention under this signal?

WHY THIS MATTERS FOR NU / NUBANK

The asset is not the PDF. The asset is the workflow.

For a Nu/Nubank reviewer, the strongest signal is not that BRAVO Lab is finished. The strongest signal is that the project demonstrates the full loop of applied financial intelligence: data handling, market intuition, derivatives reasoning, risk metrics, validation honesty, visual reporting, and executive communication.

Python infrastructure

Working project structure with scripts, processed data, report outputs, figures, tests, and reproducible execution.

Brazilian market lens

Stress is framed through B3, Brazil equity beta, FX pressure, global risk, external exposure, and volatility regimes.

Portfolio logic

The project compares passive beta, volatility income, protection, and governed switching instead of claiming one magic strategy.

Application positioning

Do not sell BRAVO Lab as an alpha engine. Sell it as evidence that you can build inspectable financial research infrastructure and convert complex Brazilian market stress into decision-grade risk material.

PORTFOLIO DECISION READ

The question is not which strategy wins. It is when each strategy is defensible.

A professional portfolio review does not need a heroic backtest. It needs a clean decision map. BRAVO Lab frames the allocation problem as a trade-off between beta exposure, income capture, protection cost, and state-dependent switching.

Passive Brazil equity

Full beta. Works when upside capture matters more than drawdown control.

Covered call

Income discipline. Useful in fragile markets, but can sell recovery too cheaply.

Collar

Left-tail protection. Defends downside, but protection cost can drag in recoveries.

BSTI policy overlay

Governed switching. Most relevant as a review rule, not a black-box signal.

The governance problem

The comparison is not about finding one strategy that always dominates. The governance problem is matching the overlay to the state: income capture in fragile markets, protection in confirmed stress, and passive beta when the evidence is too weak to justify intervention.

EXECUTIVE RISK DASHBOARD

Stress, active risk, and overlay evidence in one view

BRAVO Lab Executive Risk Dashboard

Brazilian risk, allocation, volatility, options, and stress-transmission intelligence

Latest BSTI	BSTI regime	Dominant channel
15.17	**Fragile**	**Brazil Drawdown Pressure**
Composite stress score, 0 to 100	Current transmission state	Main pressure source

Policy action	Best information ratio	Best drawdown profile
Covered Call	**Bsti Policy Overlay**	**Collar**
Current overlay rule	Highest active-risk efficiency	Least severe max drawdown

Source: BRAVO Lab processed outputs. Dashboard is generated automatically from reproducible CSV evidence files.

Why it matters

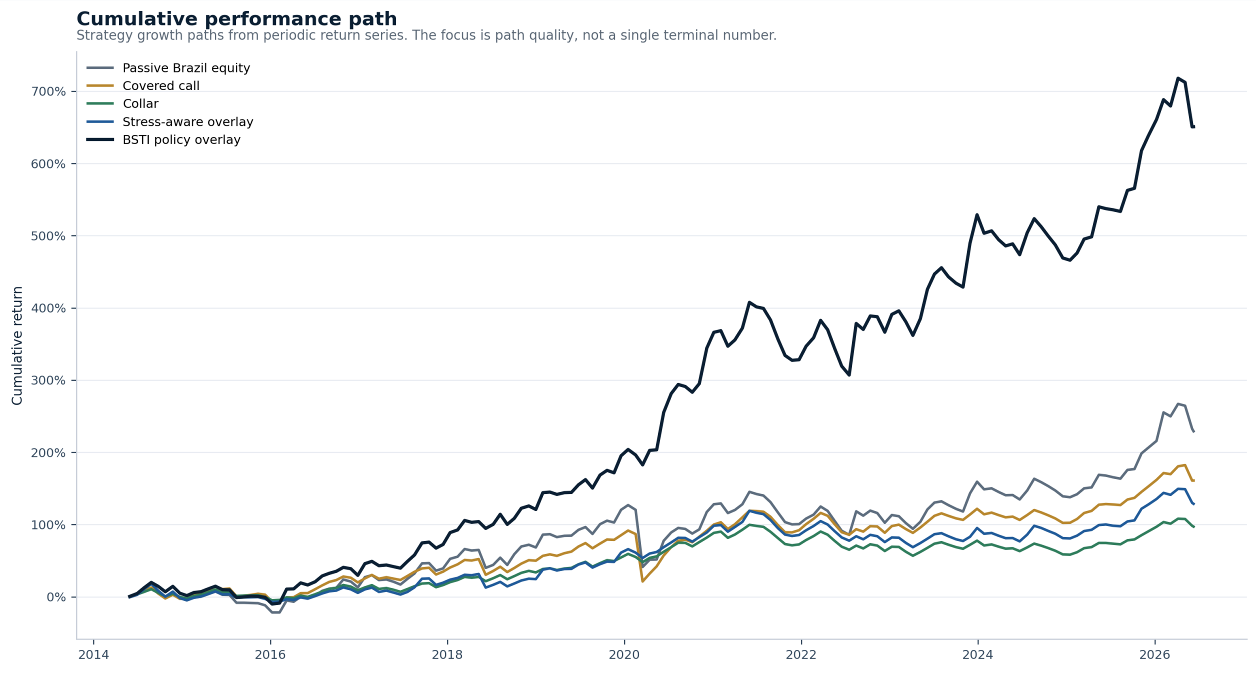
This page should let a reviewer understand the current risk state without opening the technical files.

Portfolio question

Does the current risk configuration justify changing the Brazil equity overlay, or should the portfolio remain passive?

PERFORMANCE PATH

Cumulative performance is the surface. The mandate trade-off sits underneath.



Source: BRAVO Lab overlay return table and BSTI policy overlay returns.

Why it matters

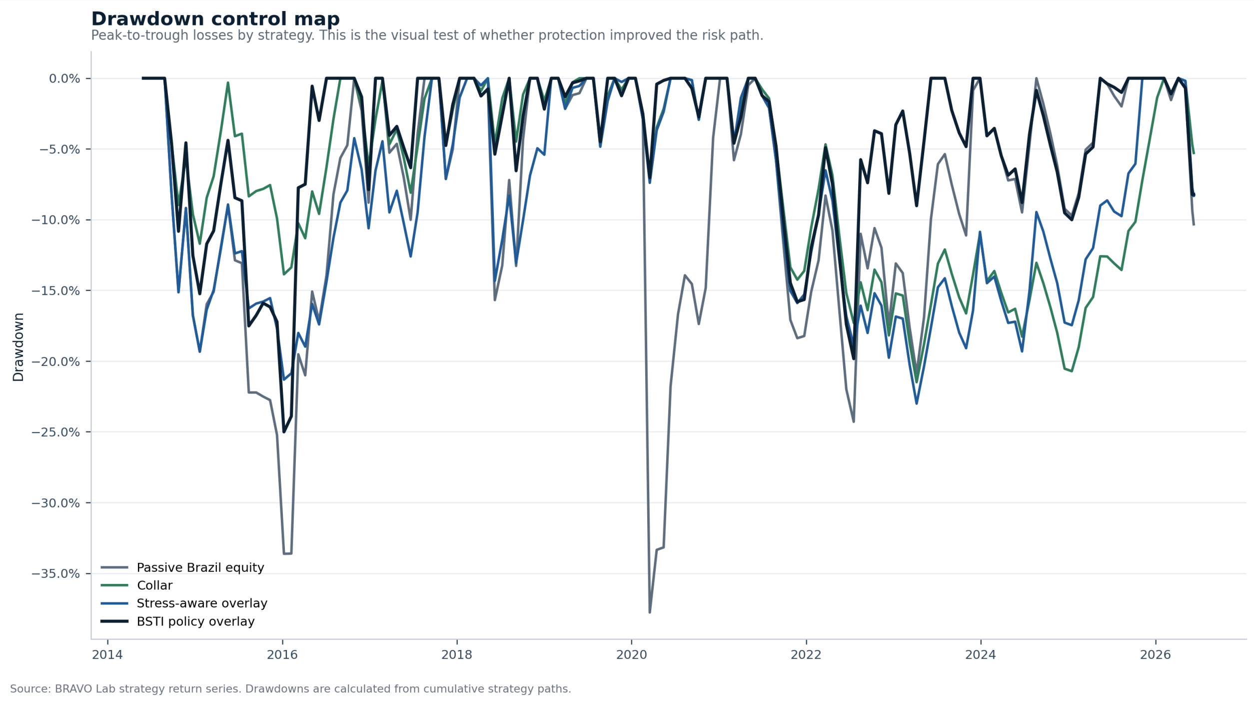
Performance alone is not the institutional question. The real question is whether the return path was achieved through acceptable drawdown, turnover, option cost, and tracking-error behavior.

Portfolio question

Would a committee accept the path if the same performance required higher implementation complexity?

DRAWDOWN CONTROL

Drawdown is where the overlay earns the right to exist



Why it matters

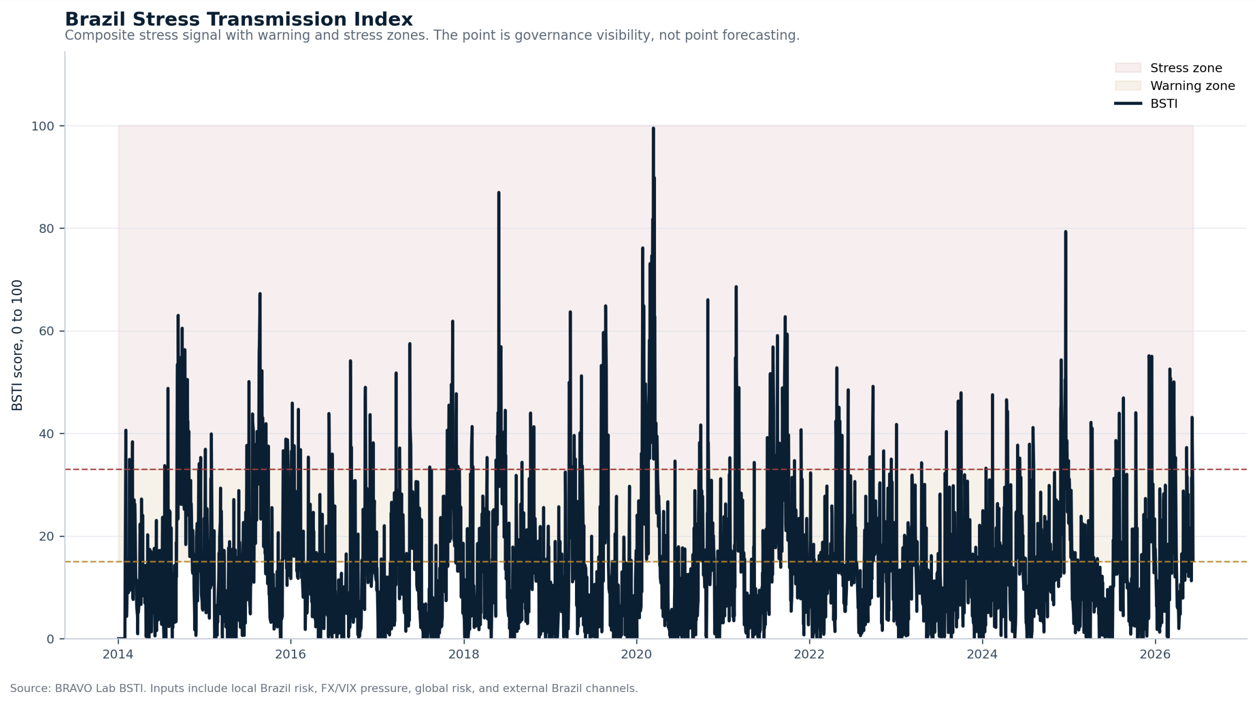
A useful overlay must show why it deserves to change the portfolio when pressure rises.

Portfolio question

Does the policy reduce drawdown enough to justify its cost, complexity, and possible upside sacrifice?

BSTI SIGNAL

Brazil Stress Transmission Index as a review trigger



Why it matters

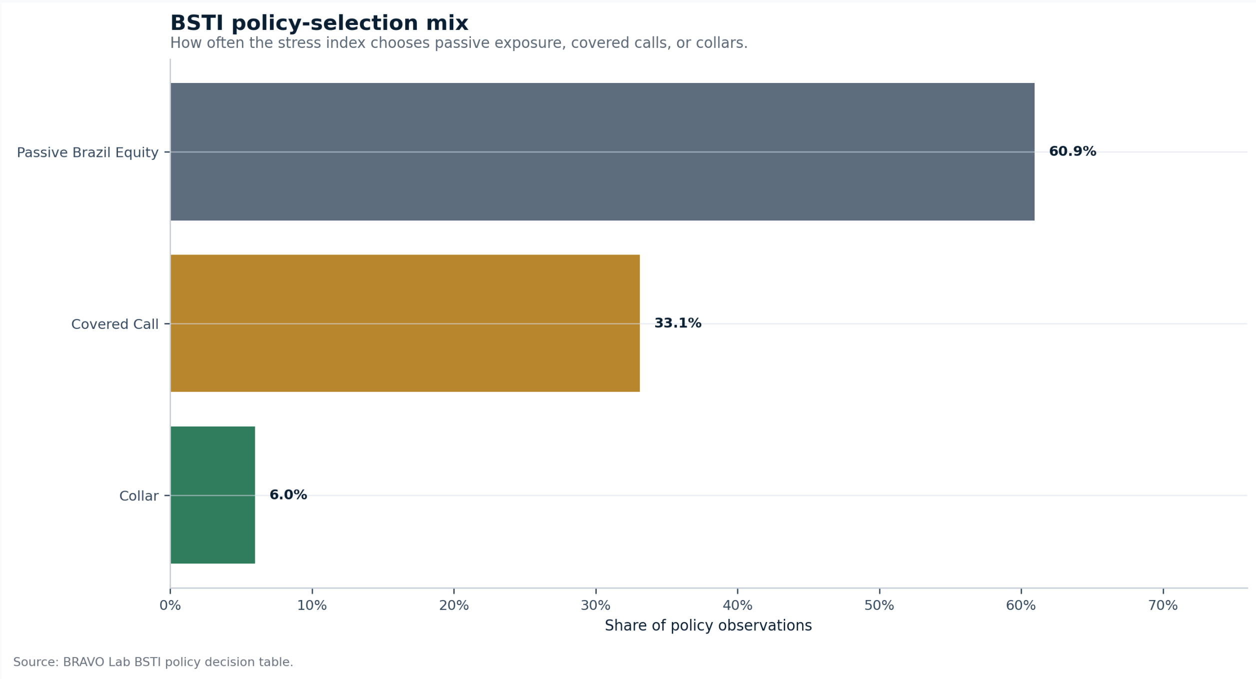
The signal should not be treated as prophecy. It should be treated as a disciplined prompt for portfolio review.

Portfolio question

At what threshold should the team move from observation to hedge review or overlay action?

POLICY SELECTION

The policy mix is the bridge between signal and action



Why it matters

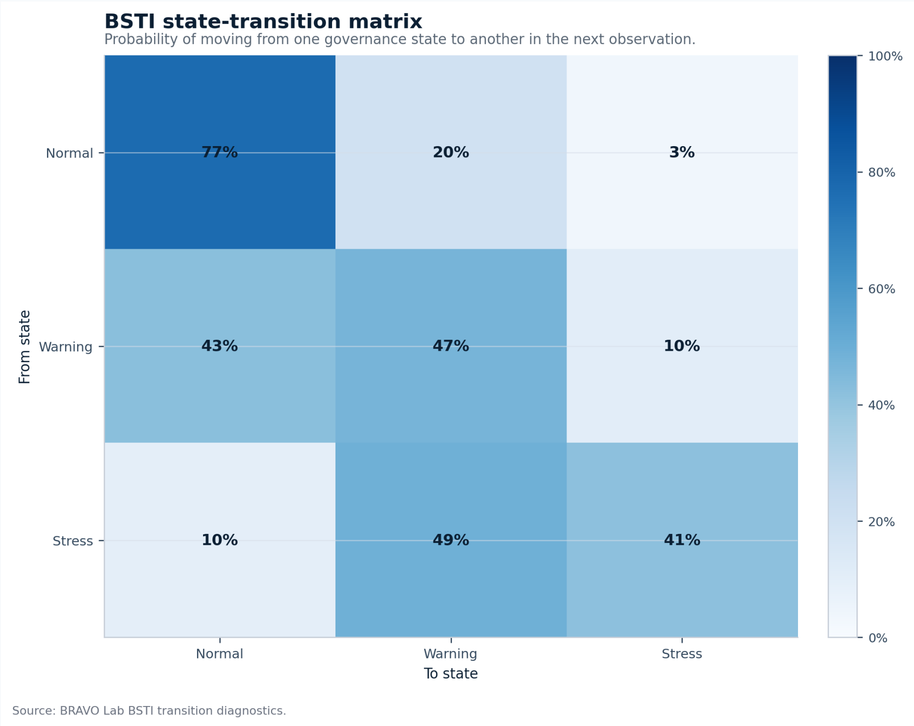
A stress index matters only if it changes behavior. The policy mix reveals whether the system is passive, defensive, or overly interventionist.

Portfolio question

Is the switching rule disciplined, or does it create unnecessary trading noise?

TRANSITION MATRIX

Regime persistence decides whether action is justified



Why it matters

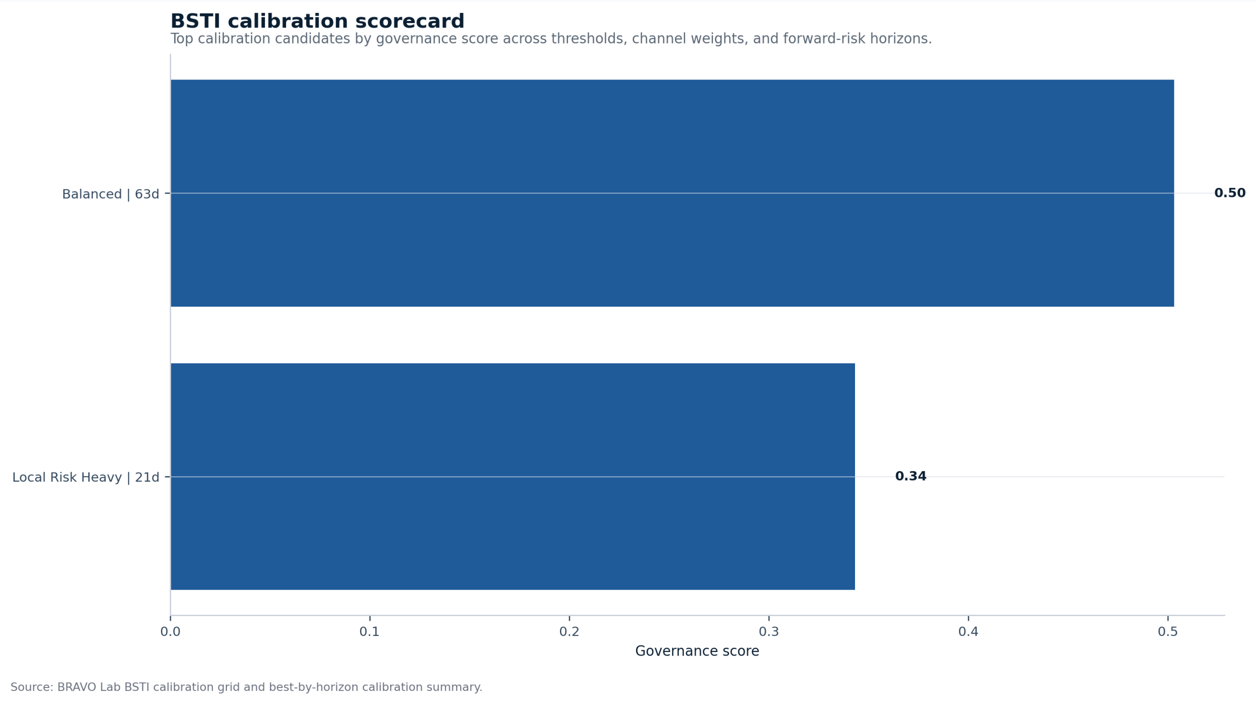
A temporary warning does not deserve the same action as a regime that tends to escalate or remain under pressure.

Portfolio question

Does the matrix support waiting, hedging, or immediate overlay action?

CALIBRATION SCORECARD

Calibration is where the prototype starts becoming auditable



Why it matters

Calibration turns a signal into a rule.
If the calibration is fragile, the
whole policy layer can become unstable.

Portfolio question

Which calibration survives out-of-sample testing, transaction costs, and real option-chain constraints?

FIRST-ORDER EFFECTS

What the signal changes immediately

- → A fragile or stress regime changes the portfolio conversation from performance maximisation to exposure discipline.
- → The covered-call overlay becomes attractive only when the committee accepts that income capture may cost upside participation.
- → A collar becomes defensible when drawdown control is more valuable than clean upside capture.
- → Tracking error must be treated as an implementation cost, not as a cosmetic metric.
- → A stress index without a decision rule is only a decorated chart.

SECOND-ORDER EFFECTS

The hidden issue is model adoption, not model output

→ A professional team will ask whether the signal survives transaction costs, taxes, liquidity constraints, and live option-chain availability.

→ Synthetic option assumptions are acceptable for prototype evidence, but not for final investment validation.

→ BSTI policy switching must be stress-tested against false positives and unnecessary trading noise.

→ A Nu/Nubank reviewer may care more about operating discipline than about the specific strategy.

→ The institutional strength is the architecture: signal, decision rule, evidence, limitation, and next validation step.

WATCH NEXT

The next build should increase trust, not decoration

→ Real B3 option-chain validation with observed strikes, maturities, bid-ask spreads, roll mechanics, and liquidity filters.

→ Transaction-cost layer including brokerage, taxes, spreads, slippage, rebalance frequency, and exercise mechanics.

→ Covariance layer using MTV-GARCH, DCC, or related volatility dependence.

→ Stress compression using CCA, Ridge CCA, or sparse CCA.

→ Wavelet layer using Airy or Generalized Morse Wavelets.

→ Nu-facing note connecting the project to applied Python, portfolio analytics, and risk infrastructure.

DEEP SWOT · RESEARCH AND PORTFOLIO VIEW

The project is strongest when it is honest about what it is and what it is not

Strengths

End-to-end research system; Brazil-specific market lens; explicit portfolio trade-offs; reproducible outputs; stress signal connected to decision language.

Weaknesses

Option assumptions are not yet fully market-validated; covariance modeling can be deeper; liquidity and tax frictions need explicit calibration.

Opportunities

Real B3 option-chain integration; MTV-GARCH or DCC volatility layer; CCA-based stress compression; wavelet decomposition; professional report automation.

Threats

Being misunderstood as a trading system; overclaiming model precision; visual design reducing credibility; insufficient validation before public release.

MODEL LIMITS AND EVIDENCE MAP

A serious prototype shows exactly where it can break

This report is intentionally explicit about its limits. A professional reviewer should see which evidence is market-based, which evidence is model-derived, and which assumptions require future validation before a live investment process could trust the system.

- → reports/front_office_memo.md — fast executive read.
- → reports/baseline_report.md — methodology and detailed evidence.
- → reports/figures/ — visual decision evidence.
- → data/processed/ — reproducible model outputs.
- → scripts/generate_institutional_pdf.py — reproducible executive report.

Final executive read

BRAVO Lab is already strong as application evidence. It should not be publicly launched until the institutional PDF, real B3 option-chain validation, and implementation-cost layer improve.

BRAVO Lab

From Brazilian market stress to portfolio-governance signal.

Research prototype · Not investment advice · Private application evidence

Created by Rodolfo Pereira